Regularize deep speaker embedding by HVAE

ADASP Formal Presentation

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Speaker verification

Objective

Decide whether two speech segments come from the same speaker or not

Method

- Front-end feature extraction: extract utterance-level speaker embeddings from audio segments, e.g. i-vector, **x-vector**.
- Back-end scoring: compute the similarity score between two embeddings and decide against a preset threshold.

Scoring methods

- cosine similarity: $\frac{x_1^T x_2}{||x_1||||x_2||}$
- probabilistic linear discriminant analysis:

Suppose the speaker embedding is generated from the following model

$$P(\mathbf{y}) = \mathcal{N}(\mathbf{y} \mid \mathbf{m}, \Phi_b)$$
$$P(\mathbf{x} \mid \mathbf{y}) = \mathcal{N}(\mathbf{x} \mid \mathbf{y}, \Phi_w)$$

Then by hypothesis test, we compute the similarity as the loglike-ratio:

$$\begin{aligned}\log R(\mathbf{x}_1, \mathbf{x}_2) &= \log \frac{P\left(\mathbf{x}_1, \mathbf{x}_2 | \mathcal{T}, \theta\right)}{P\left(\mathbf{x}_1, \mathbf{x}_2 | \mathcal{N}, \theta\right)} = \log \frac{\int p(x_1|y)p(x_2|y)p(y)dy}{\int p(x_1|y)p(y)dy \int p(x_2|y)p(y)dy} \\ &= (x_1 + x_2)^T G (x_1 + x_2) - x_1^T H x_1 - x_2^T H x_2\end{aligned}$$

where $G = \Phi_W^{-1}(\Phi_B^{-1} + 2\Phi_W^{-1})\Phi_W^{-1}, H = \Phi_W^{-1}(\Phi_B^{-1} + \Phi_W^{-1})\Phi_W^{-1}$

Decision

Decision by threshold

Given a threshold θ, we decide pairs with scoring s

$$\begin{cases} s \geq \theta, & \text{same speaker} \\ s < \theta, & \text{different speakers} \end{cases} \tag{1}$$

Two types of error in decision

- False Reject: If two segments from a same speaker are decided to be from different speakers
- False Accept: If two segments from different speakers are decided to be from a same speaker.

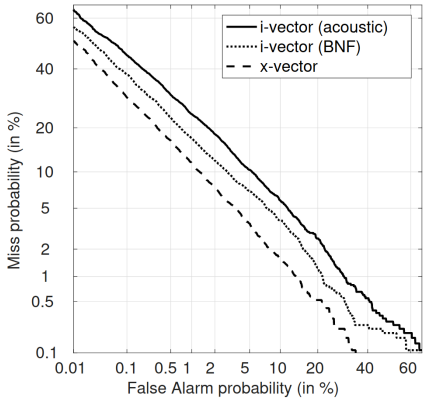
Therefore if we have a large threshold, the system tends to be very conservative, which increases FR error but lowers FA and vice versa.

In practice, the setting of threshold depends on the application: "security-first" or "convenience-first".

Evaluation

Regardless of pratical context, to evaluate the verification system, we have two typical methods:

- Draw the FA-FR curve: draw the (FA, FR) curve parametrized by the threshold.



- Report the Equal Error Rate (i.e. the cross point of the curve and $y = x$)

Issue

Before deep speaker embeddings (x-vectors), PLDA is the major scoring method for the previous SOTA embedding(i-vector). However, when deep speaker embeddings come to play, cosine similarity is better than PLDA.

Guess

Unlike i-vector which is extracted from a GMM, x-vector which is extracted from a neural network (one or two layers before the classification head) follows badly the Gaussianity assumption of PLDA.

Attemps

We try to regularize the deep speaker embedding to make it appropriate for PLDA.

Previous works

1. Gaussian-constrained xvector training: regularization during training of feature extractor
2. regularization of xvector by VAE
3. regularization of xvector by Normalizing flow

Shortbacks

- Method 1 needs to train everything from scratch
- Method 2 only constrain the speaker prior $p(y)$ to Gaussian but not the conditional $p(x|y)$
- Method 3 only constrain the speaker marginal $p(x)$

We propose an HVAE to contrain both the speaker prior and speaker conditional to Gaussian.

Extend PLDA to non-linear analog

We hypothesize that the PLDA works no more due to the non-linearity of neural embeddings.

Formulation

Instead of Probablistic Linear discriminant analysis, we replace the linear dependency of utterance to speaker by non-linear one, i.e.

$$\mathbf{y}_i \sim p(\mathbf{y}) = \mathcal{N}(\mathbf{y}; \mathbf{0}, \mathbf{I}), \tag{2}$$

$$\mathbf{z}_{ij}|\mathbf{y}_i = \mathbf{y} \sim p(\mathbf{z}|\mathbf{y}) = \mathcal{N}(\mathbf{z}; f_{\boldsymbol{\theta}}(\mathbf{y}), \mathbf{I}), \tag{3}$$

$$\mathbf{x}_{ij} = g_{\boldsymbol{\theta}}(\mathbf{z}_{ij}), \tag{4}$$

where f_θ and g_θ is generally non-linear and we keep the covariances independent of speakers and normalized.

Solving the model by VI

The ELBO function for the observed N_i utterances $\mathbf{X}_i = \{\mathbf{x}_{ij}\}_{j=1}^{N_i}$ with hidden utterances $\mathbf{Z}_i = \{\mathbf{z}_{ij}\}_{j=1}^{N_i}$ from speaker $\mathbf{y}_i$ of the model:

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{X}) := \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{y}, \mathbf{Z}|\mathbf{X})}[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{X}, \mathbf{Z}, \mathbf{y})}{q_{\boldsymbol{\phi}}(\mathbf{y}, \mathbf{Z}|\mathbf{X})}], \tag{5}$$

$$= \mathbb{E}_{q(\mathbf{y}, \mathbf{Z}|\mathbf{X})}\left[\log p(\mathbf{X}|\mathbf{Z})\right] - \mathrm{KL}(q(\mathbf{y}, \mathbf{Z}|\mathbf{X})||p(\mathbf{y}, \mathbf{Z})), \tag{6}$$

By top-down inference: $q(\mathbf{y}, \mathbf{Z}|\mathbf{X}) = q(\mathbf{y}|\mathbf{X})q(\mathbf{Z}|\mathbf{y}, \mathbf{X})$

$$\begin{aligned} &\mathrm{KL}(q(\mathbf{y}, \mathbf{Z}|\mathbf{X})||p(\mathbf{y}, \mathbf{Z})) = \\ &\mathrm{KL}(q(\mathbf{y}|\mathbf{X})||p(\mathbf{y})) + \mathbb{E}_{q(\mathbf{y}|\mathbf{X})}[\mathrm{KL}(q(\mathbf{Z}|\mathbf{y}, \mathbf{X})||p(\mathbf{Z}|\mathbf{y}))]. \end{aligned} \tag{7}$$

Solving the model by VI

Factorize the speaker conditional KL by the indepency between utterances from the same speaker.

$$\mathrm{KL}(q(\mathbf{Z}|\mathbf{y},\mathbf{X})||p(\mathbf{Z}|\mathbf{y})) = \sum_{j=1}^{N_i} \mathrm{KL}(q(\mathbf{z}_j|\mathbf{y},\mathbf{X})||p(\mathbf{z}_j|\mathbf{y})) \tag{8}$$

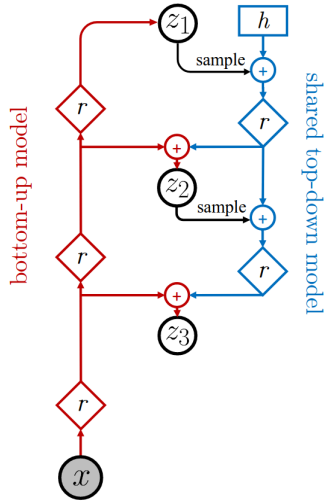
The ELBO for the whole dataset of all speakers is the sum of individual ELBOs, by the independency between speakers:

$$\log p_{\boldsymbol{\theta}}(\mathbf{X}) = \sum_{i=1}^{S} \log p_{\boldsymbol{\theta}}(\mathbf{X}_i) \geq \mathcal{L}(\boldsymbol{\theta},\boldsymbol{\phi};\mathbf{X}) = \sum_{i=1}^{S} \mathcal{L}(\boldsymbol{\theta},\boldsymbol{\phi};\mathbf{X}_i) \tag{9}$$

HVAE

The above objective can be modeled with a hierarchical VAE, since we have two KL-terms: one for speaker prior and one for speaker conditional.

We borrow the implementation from NVAE, adapt it with two stochatic layers and replace the convolution layers to fully-connected ones.



Scoring on hidden space

Once we get our regularized embeddings y, we can perform hypothesis test as follows:

$$
\begin{aligned}
R(\mathbf{x}_1, \mathbf{x}_2) &= \frac{P\left(\mathbf{x}_1, \mathbf{x}_2 | \mathcal{T}, \theta\right)}{P\left(\mathbf{x}_1, \mathbf{x}_2 | \mathcal{N}, \theta\right)} = \frac{\int p(x_1|y)p(x_2|y)p(y)dy}{p(x_1)p(x_2)} \\
&= \int \frac{p(x_1, y)p(x_2, y)}{p(x_1)p(x_2)p(y)}dy = \int \frac{p(y|x_1)p(y|x_2)}{p(y)}dy \\
&\approx \int \frac{q(y|x_1)q(y|x_2)}{q(y)}dy = x_1^T x_2 + const.
\end{aligned}
$$

Such simplification of score is due to our assumptions on the identity covariances. If we also perform the common length-normalization trick, we conclude that the scoring of our regularized embedding reduced to the cosine similarity on hidden space Y.

Experiment and Results

We use two extractors in experiments: one come from Sidiar(weaker), and one come from Wenet(Stronger), both of them are trained on Voxceleb.

embedding	Vox1o(Sidiar/Wenet)	CN-Celeb(Sidiar/Wenet)
x(cos)	2.1/0.9703	19.42/11.0673
x(plda)	2.7/2.2	21.58/16
y(cos)	3.1/2.6	18.38/13.9847
z(cos)	-/1.696	-/13.8439

On weaker checkpoint, the regularization seems to benefit the out-of-domain data (CN-Celeb), but when we switch to stronger checkpoint, we no more benefit from such regularization.

Future improvements

1. Change the prior from Gaussian to vMF.
2. Adding supervised labels in the HVAE loss.

END